Assumptions, diagnostics, and multicollinearity

Last time: Variance inflation factors

Motivation: High multicollinearity means a predictor can be estimated well by some combination of other predictors

Variance inflation factor: Let X_i be a predictor in the model, and R_i^2 the coefficient of determination for a model estimating X_i from the other predictors. The *variance inflation factor* of X_i is

$$VIF_i = \frac{1}{1 - R_i^2}$$

- Higher VIF means more multicollinearity
- Rule of thumb: worried when $VIF_i > 5$ (i.e. $R_i^2 > 0.8$)

Last time: Class activity

- $\widehat{\text{Hgt97}} = 40.59 + 1.10\,\text{Hgt96} \qquad R^2_{adj} =$

Class activity

- $\widehat{\text{Hgt97}} = 40.59 + 1.10\,\text{Hgt96} \qquad R^2_{adj} = 0.949$
- $\widehat{\text{Hgt97}} = 40.24 - 1.26\,\text{Diam96} + 1.12\,\text{Hgt96} \qquad R^2_{adj} =$

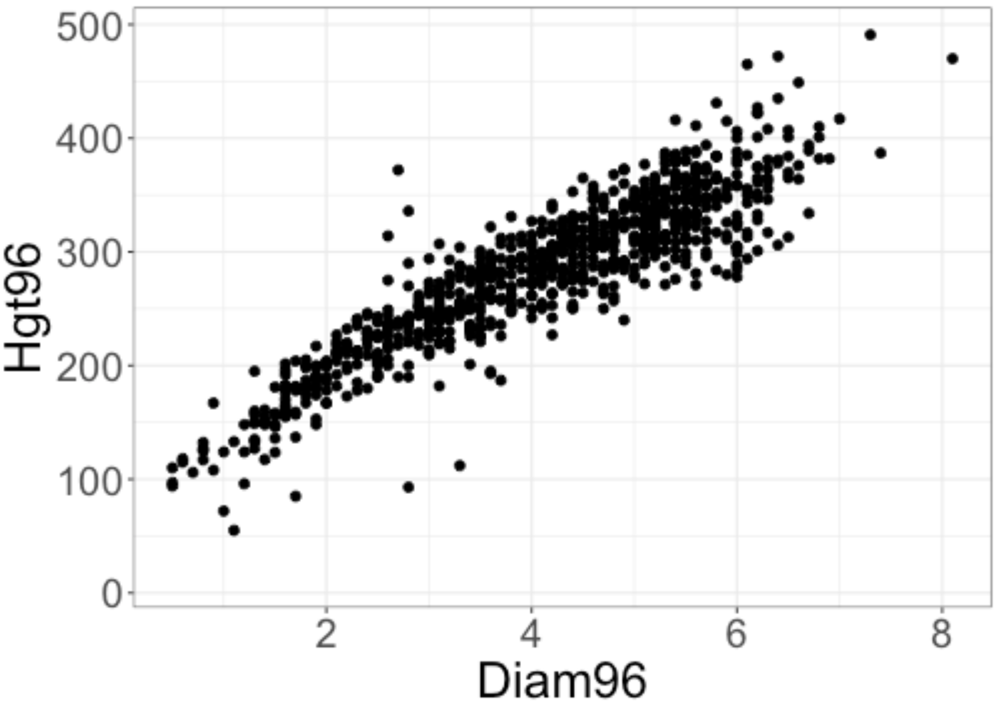
Class activity

- $\widehat{\text{Hgt97}} = 40.59 + 1.10\ \text{Hgt96} \qquad R^2_{adj} = 0.949$
- $$\widehat{\text{Hgt97}} = 40.24 - 1.26\ \text{Diam96} + 1.12\ \text{Hgt96} \qquad R^2_{adj} = 0.942$$

> Why doesn't R^2_{adj} increase when I add diameter to the model?

Class activity

> Why doesn't R^2_{adj} increase when I add diameter to the model?



Because diameter and height in 1996 are so correlated.

For two quantitative variables:

$R^2 = r^2$

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26 \text{ Diam96} + 1.12 \text{ Hgt96}$$

What is the variance inflation factor for Diam96?

$$VIF = \frac{1}{1 - R^2}$$

R^2 : coefficient of determination for predicting Diam96 from Hgt96

$R^2 = 0.813$

$$\Rightarrow VIF = \frac{1}{1 - 0.813} = 5.35$$

VIF for Hgt96 $= 5.35$

Class activity

$$\widehat{\text{Hgt97}} = 40.24 - 1.26\ \text{Diam96} + 1.12\ \text{Hgt96}$$

> What is the variance inflation factor for Diam96?

```
m1 <- lm(Diam96 ~ Hgt96, data = Pines)
summary(m1)$r.squared
```

```
## [1] 0.8134518
```

$$VIF = \frac{1}{1 - 0.813} = 5.35$$

Why is multicollinearity a problem?

- Inflates variability of estimated coefficients
- Interpretation of individual terms is difficult

$$\widehat{\sqrt{\text{MDs}}} = 3.58 + 2.58 \text{ Hospitals} + 0.012 \text{ Beds}$$

Usual interpretation: Holding Beds fixed, an increase of 1 hospital is associated with an increase of 2.58 units in $\sqrt{\text{MDs}}$ (on average)

Problem: When Beds and Hospitals are highly correlated, doesn't make sense to fix one and change the other. (Can't really change the number of hospitals without changing the number of beds)

Example data

- `MED_EARNINGS`: median earnings for graduates from each university (response)
- `CONTROL`: whether the school is public or private
- `SATVRMID`: midpoint of SAT critical reading scores of students attending the school
- `SATMTMID`: midpoint of SAT math scores
- `SATWRMID`: midpoint of SAT writing scores
- `ACTCMMID`: midpoint of the ACT cumulative scores
- `ACTENMID`: midpoint of ACT English scores
- `ACTMTMID`: midpoint of ACT math scores
- + others...

> Which variables might exhibit multicollinearity?

Example data

- `MED_EARNINGS`: median earnings for graduates from each university
- `CONTROL`: whether the school is public or private
- `SATVRMID`: midpoint of SAT critical reading scores of students attending the school
- `SATMTMID`: midpoint of SAT math scores
- `SATWRMID`: midpoint of SAT writing scores
- `ACTCMMID`: midpoint of the ACT cumulative scores
- `ACTENMID`: midpoint of ACT English scores
- `ACTMTMID`: midpoint of ACT math scores
- + others...

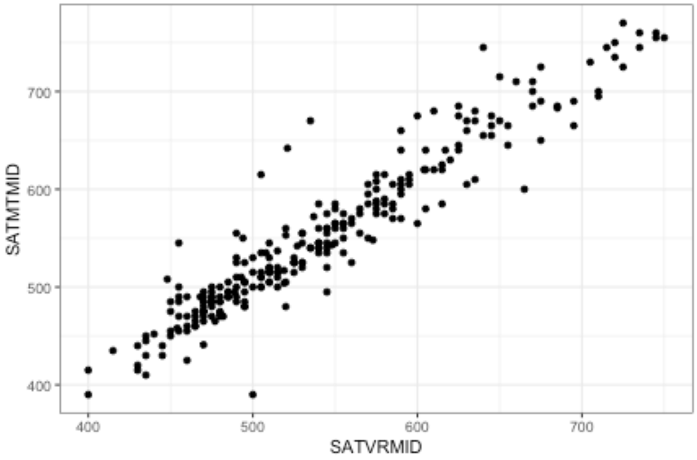
There is probably multicollinearity with all the test score variables

Example data

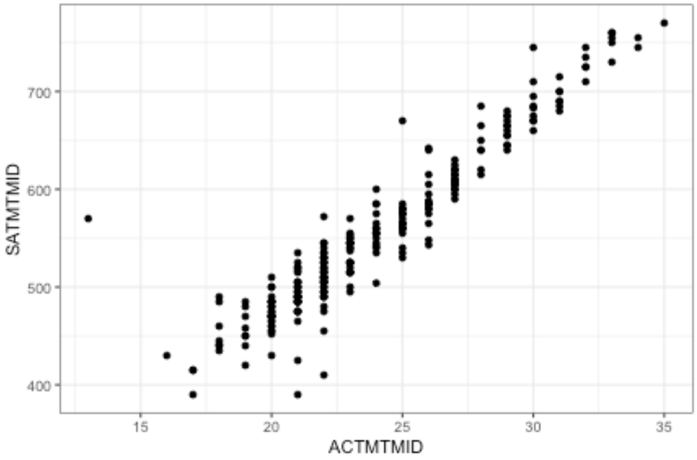
- `MED_EARNINGS`: median earnings for graduates from each university
- `CONTROL`: whether the school is public or private
- `SATVRMID`: midpoint of SAT critical reading scores of students attending the school
- `SATMTMID`: midpoint of SAT math scores
- `SATWRMID`: midpoint of SAT writing scores
- `ACTCMMID`: midpoint of the ACT cumulative scores
- `ACTENMID`: midpoint of ACT English scores
- `ACTMTMID`: midpoint of ACT math scores
- + others...

> How can we visually investigate correlated predictors?

Example data



$r = 0.95$



$r = 0.94$

VIFs in R

Fortunately, R makes it easy to calculate VIFs:

```
library(car)
m1 <- lm(MED_EARNINGS ~ SATVRMID + SATMTMID +
           SATWRMID + ACTCMMID + ACTENMID +
           ACTMTMID + ACTWRMID,
         data = scorecard)
vif(m1)
```

```
## SATVRMID SATMTMID SATWRMID ACTCMMID ACTENMID ACTMTMID AC
## 29.78240 17.69560 27.89062 27.68832 21.90292 17.26265  1
```

- The `vif` function (from the `car` package) calculates a VIF for each predictor
- The variance inflation factor for SAT verbal score is 29.78, e.g.

Handling multicollinearity

We have a few options for handling multicollinearity:

- Remove variables
  - Example: just pick one of the test score variables, and discard the rest, since they are so highly correlated

Handling multicollinearity

We have a few options for handling multicollinearity:

- Remove variables
  - Example: just pick one of the test score variables, and discard the rest, since they are so highly correlated
- Create a new variable
  - Example: take an average or sum of the test score variables, instead of using all of them individually

Handling multicollinearity

We have a few options for handling multicollinearity:

- Remove variables
  - Example: just pick one of the test score variables, and discard the rest, since they are so highly correlated
- Create a new variable
  - Example: take an average or sum of the test score variables, instead of using all of them individually
- Ignore it
  - If we don't care about interpreting coefficients or doing inference, multicollinearity doesn't matter as much